ESTADÍSTICA Y DISEÑO EXPERIMENTAL

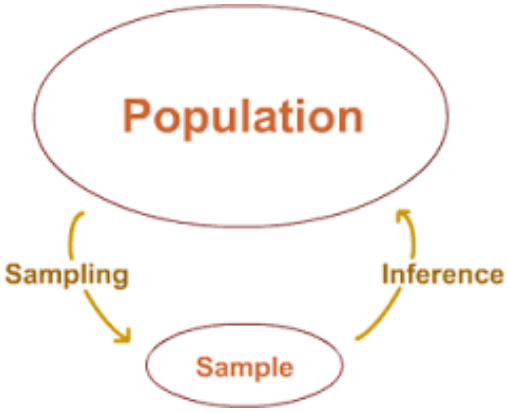
EXPERMIENTAL DESIGN

Rafael Valdés Mas
rvaldesmas@unav.es

INTRODUCTION

Experimentation:

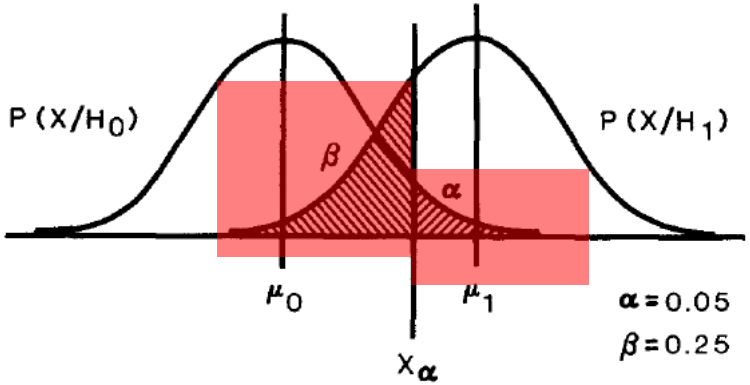
- Experimentation is an essential part in research
- Test hypothesis and assess the relationship between different variables
- We can evaluate whether differences can be randomly obtained or are certainly significant through **statistical hypothesis testing**.
- We have plenty of tests to assess our hypotheses (t-test, f-test, ANOVA, binomial test, hypergeometric test,…)
- The success of this statistical validation tools depends on an appropriate **experimental design**.



POWER AND SAMPLE SIZE

- Important decision in experimental design**: sample size** (number of experiments)
- **Clinical trials.** How many individuals will I need to study?
- Number of sensors, test tubes, animals…
- Consider a mean difference test and their distributions below:

  H_0: $N(\mu_0,\sigma_0)$ and **H_1: $N(\mu_1,\sigma_1)$, $\mu_1>\mu_0$**
- **Type I errors** (false positive, α) **and Type II errors** (false negative, β)



Power= 1- β
Probability of capturing a true positive

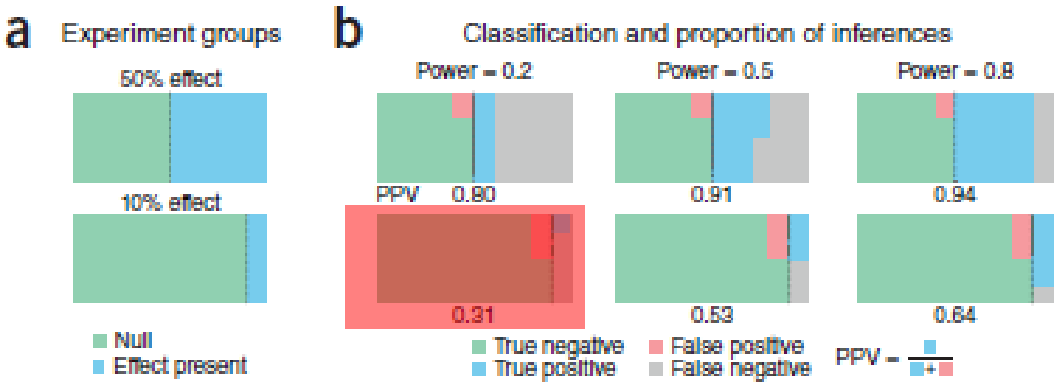
SCRATCH DETECTION

SCREEN SCRATCH DETECTION

```
# Clean the workspace
rm(list = ls())
graphics.off()

# Input data
mu0 <- 0; mu1 <- 0.; sigma <- 1; n <- 9
delta <- (mu1 - mu0) * sqrt(n) / sigma

# alpha
alpha <- 0.05
z_alpha <- qnorm(1 - alpha)

# beta
beta <- pnorm(z_alpha - delta)
power <- 1 - beta
```

POWER AND SAMPLE SIZE

- **Screening (exploratory) analysis**



Krzywinski and Altman, 2013

PPV = (1- β)*effect/((1- β)*effect + α*no_effect)

PPV = 0.2*0.1/(0.2*0.1 + 0.05*0.9)= 0.31

Positive Predictive Value

PPV = proportion of correctly predicted positives

SCREEN SCRATCH DETECTION

```
# Clean the workspace
rm(list = ls())
graphics.off()

# Input data
mu0 <- 0; mu1 <- 0.; sigma <- 1; n <- 9
delta <- (mu1 - mu0) * sqrt(n) / sigma

# alpha
alpha <- 0.05
z_alpha <- qnorm(1 - alpha)

# beta
beta <- pnorm(z_alpha - delta)
power <- 1 – beta

# PPV
penetrance = 0.01
PPV = (1-beta) * penetrance / ( (1-beta) * penetrance + alpha * (1 - penetrance) )
```

SCREEN SCRATCH DETECTION

plot curve of PPV based on different penetrances

do the same for evolution of Power based on sample size

POWER AND SAMPLE SIZE

Studies with inadequate power are a waste of research resources and arguably unethical when subjects are exposed to potentially harmful or inferior experimental conditions. Addressing this shortcoming is a priority—the Nature Publishing Group checklist for statistics and methods (http://www.nature.com/authors/policies/checklist.pdf) includes as the first question: "How was the sample size chosen to ensure adequate power to detect a pre-specified effect size?" Here we discuss inference errors and power to help you answer this question. We'll focus on how the sensitivity and specificity of an experiment can be balanced (and kept high) and how increasing sample size can help achieve sufficient power.

Krzywinski and Altman, 2013, Nature Methods

POWER AND SAMPLE SIZE

nature research

Corresponding author(s): Eran Elinav

Last updated by author(s): May 6, 2021

nature research | reporting summary

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used to collect the data.
Data analysis	No custom code was used in the analysis. We used the following software: Fastp (0.20.1) (https://github.com/OpenGene/fastp), Bowtie2 (2.4.1) (https://github.com/BenLangmead/bowtie2), Seqtk (1.3-r114) (https://github.com/lh3/seqtk), Kraken2 (2.0.9) (https://github.com/DerrickWood/kraken2), Bracken (2.5.3) (https://github.com/jenniferlu717/Bracken), ShortBRED (0.9.5) (https://github.com/biobakery/shortbred), DeepARG (2.0) (https://github.com/gaarangoa/deeparg2.0), ARGs-OAP (2.0) (https://github.com/biofuture/Ublastx_stageone),, GROOT (1.1.2) (https://github.com/will-rowe/groot), Blastp (2.10.1+) (https://blast.ncbi.nlm.nih.gov/Blast.cgi?CMD=Web&PAGE_TYPE=BlastDocs&DOC_TYPE=Download), sraX (1.5) (https://github.com/lgpdevtools/sraX), Spades (3.14.1) (https://github.com/ablab/spades), GraphPad Prism (9.1.0) (https://www.graphpad.com/scientific-software/prism/), R (4.0.4), Vegan R package (2.5-7) (https://rdrr.io/cran/vegan/man/vegan-package.html), Phyloseq R package (1.32.0) (http://www.bioconductor.org/packages/release/bioc/html/phyloseq.html). The following databases were used in the analyses: H. sapiens, GRCh37 (http://hgdownload.soe.ucsc.edu/goldenPath/hg19/bigZips/), Mus Musculus, C57BL_6nJ (https://www.ncbi.nlm.nih.gov/genome/52?genome_assembly_id=422183), Genome Taxonomy Database r89 (https://data.gtdb.ecogenomic.org/releases/), Comprehensive Antibiotic Resistance Database (CARD) (1.05) (https://card.mcmaster.ca/download), Database of mobile genetic elements (transposases, integrases, recombinases and integrons), curated by NanoARG (1.0) (https://bench.cs.vt.edu/nanoarg/#/home), ARGminer database (1.1.1) (https://bench.cs.vt.edu/argminer/#/database).

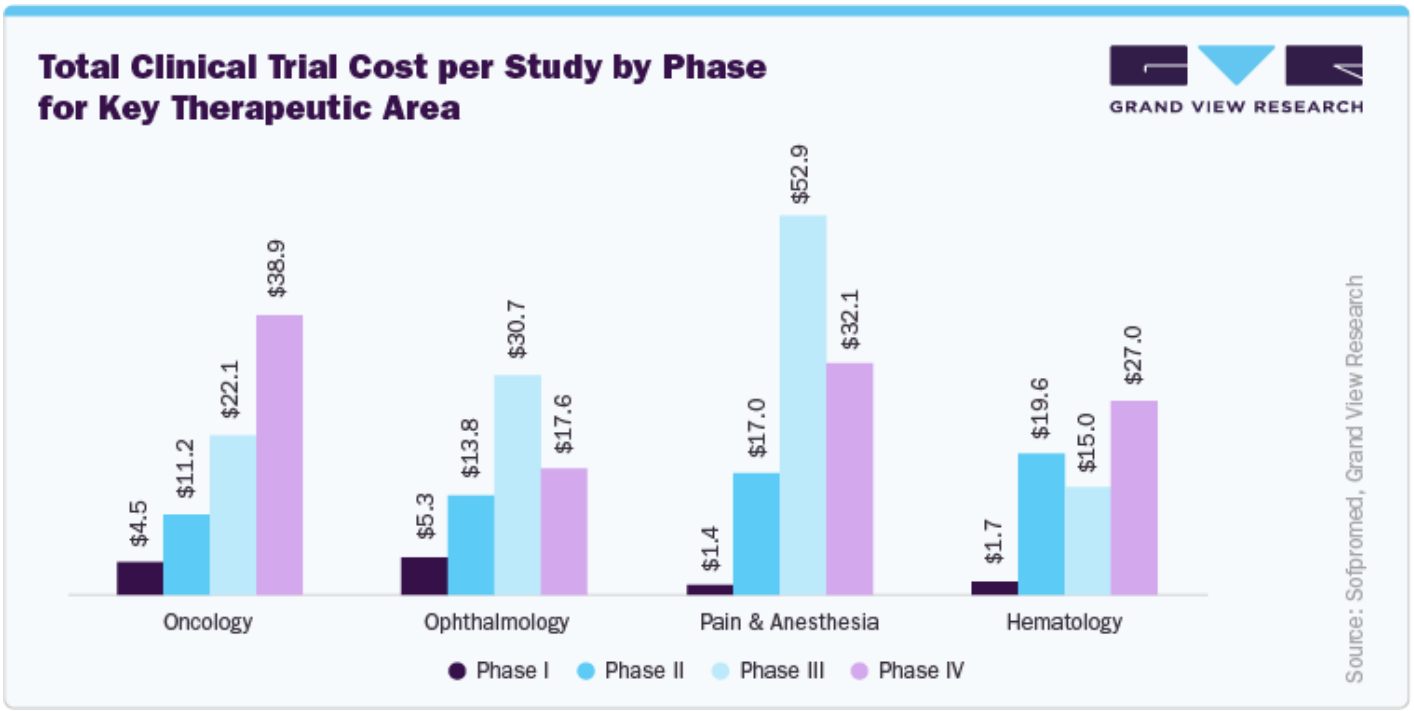
For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

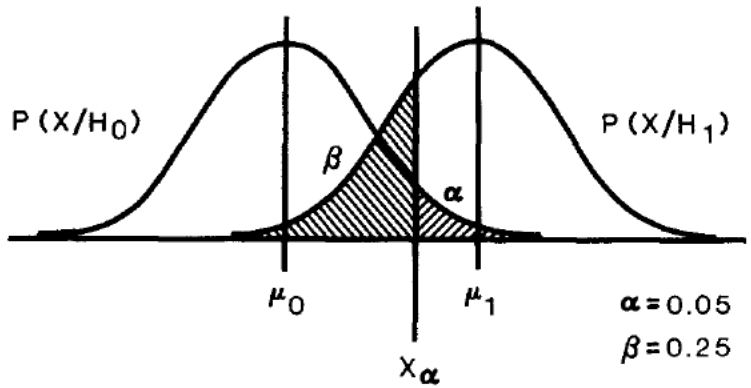
Sample size	This work analyzes data from a published clinical trial without adding new participants to the original cohort. No sample size calculations were performed for this work.
Data exclusions	No data were excluded from the analysis.
Replication	The observation that stool samples do not reflect the gastrointestinal resistome in antibiotics-naive individuals was successfully replicated in antibiotics-treated individuals. Probiotics-associated expansion of the gastrointestinal resistome in antibiotics-treated humans was successfully replicated in mice. Reanalysis of additional publicly available data provided some support to the importance of person-specific resistome analysis and direct sampling.
Randomization	This work analyzes data from a published clinical trial without adding new participants to the original cohort. No randomization was performed for this work.
Blinding	This work analyzes data from a published clinical trial without adding new participants to the original cohort. Data analyses were performed on barcoded samples without group identification.

POWER AND SAMPLE SIZE



Millions of dollars

POWER AND SAMPLE SIZE



$$x_\alpha = \mu_0 + z_\alpha \Sigma_0$$
$$x_\alpha = \mu_1 - z_\beta \Sigma_1$$

$$\boxed{|\mu_1 - \mu_0| = z_\alpha \Sigma_0 + z_\beta \Sigma_1}$$

$$|\mu_1 - \mu_0| = Z_\alpha \Sigma_0 + Z_\beta \Sigma_1$$

Variance depends on n (sample size)

1. What *sample size* is required to ensure power $1 - \beta$ of detecting a relevant difference μ_1?
2. What is the *power* (Z_β) of the experiment in detecting a relevant difference μ_1 when a specific sample size N is employed?
3. What *difference* μ_1 can be detected with power $1 - \beta$ if the experiment is conducted with a specified sample size N?

Lachin, 1981

POWER AND SAMPLE SIZE

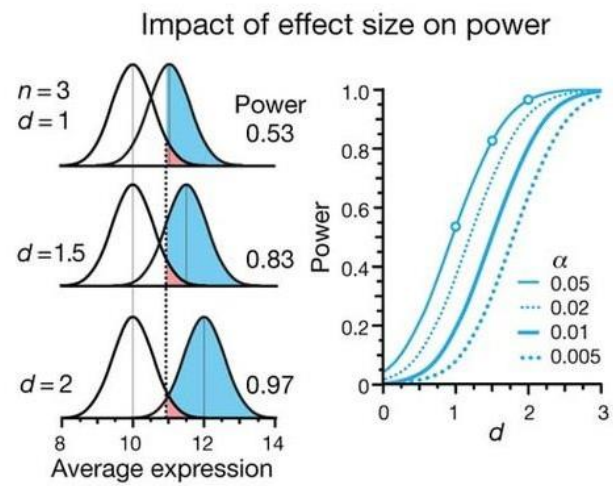
Example:

- **Two-sample mean difference test.**
- **H_0: $\mu_1-\mu_0=0$, H_1: $\mu_1-\mu_0>0$**
- Sample size (n) is the same for both groups.
- Variance is known and the same for both groups:

$$\sqrt{n}\cdot|\mu_1-\mu_0| = \sqrt{2}\,\sigma(z_\alpha + z_\beta)$$

Effect Size:

- $d=|\mu_1-\mu_0|/\sigma$
- **Z-score** that quantifies the difference of effect between null and alternative hypothesis.
- It is a measure of overlap between null and alternative hypothesis.



Krzywinski and Altman, 2013, Nature Methods

TWO-SAMPLE MEAN DIFFERENCE TEST

Step by step on the board:

$$N = \frac{4\sigma^2(z_\alpha + z_\beta)^2}{|\mu_1 - \mu_0|^2}$$

EXERCISE 1 - POWER AND SAMPLE SIZE

a) Assuming $\boldsymbol{\alpha}$=0.05; $\boldsymbol{\beta}$=0.2; $\boldsymbol{\mu_1 - \mu_0}$=3; $\boldsymbol{\sigma}$=10, determine the sample size (**N**) if we aim to conduct a mean difference test between two independent samples of equal size (variance is known).

b) Download the R package "**pwr**" and compare the results obtained in question **a** using ***pwr.t.test*** function

$$\sqrt{n}\cdot|\mu_1-\mu_0| = \sqrt{2}\,\sigma(t_{n-2,\,\alpha} + t_{n-2,\,\beta})$$

Iterative procedure: $|\mu_1-\mu_0| = \sigma_0 \cdot t_{n-2,\,\alpha} + \sigma_1 \cdot t_{n-2,\,\beta}$

a) Evaluate the sample size for $\boldsymbol{\sigma}$ in the interval [5,15]

EXERCISE 1A- POWER AND SAMPLE SIZE

```
# Clean the workspace
rm(list = ls())
graphics.off()

# Input data
u1 <- 3 ; u0 <- 0; sigma <- 10; alpha <- 0.05; beta <- 0.2

# Effect size
d <- abs(u1-u0)/sigma

# Power
power <- 1-beta

# Calculate N using the formula
zalpha <- abs(qnorm(alpha))
zbeta  <- abs(qnorm(beta))
N <- 4*((zalpha + zbeta)^2)/d^2
```

EXERCISE 1B- POWER AND SAMPLE SIZE

```
# Calculate N using pwr package

# Load package
library(pwr)

# Apply function for t.test
P <- pwr.t.test(d=d,sig.level=alpha,
                power=power,type="two.sample",alternative="greater")
Npwr <- 2*P$n
Npwr
```

EXERCISE 1C- POWER AND SAMPLE SIZE

Plot N for different values of sigma (Exercise 1a)

To do in class

EXERCISE 2- ANOVA AND SAMPLE SIZE

a) Generate **N=10** random values for 3 different groups, which are normally distributed as follows: **N(μ_1,1), N(μ_2,1)** and **N(μ_3,1)** , **μ_1**=0.8; **μ_2**=1; **μ_3**=1.1. Use functions ***aov.*** The function ***gl*** will be useful to generate the different classes.

b) Find the sample size ***N*** required for the test to be sufficiently powerful to capture the differences using ***pwr*** package.

$$f = \frac{\sqrt{\sum(u_i - u)^2/3}}{\sigma}$$

EXERCISE 2A- ANOVA AND SAMPLE SIZE

```
# Clean the workspace
rm(list = ls()); graphics.off(); set.seed(500)

# Input Data
n <- 10; u1 <-0.8; u2 <-1; u3 <-1.1; sigma<-1; l <- 3

# Generate factors
f <- c("T1", "T2", "T3")
tm <- gl(l, 1,  n*k,  factor (f) )

# Generate output values
t1 <- rnorm(n, u1,sigma)
t2 <- rnorm(n, u2,sigma)
t3 <- rnorm(n, u3,sigma)
y <- matrix (c(t1,t2, t3), nrow=n, ncol=l)
y <- c(t(y))

# Anova
av <-  aov(y ~ tm, projections=TRUE)
summary(av)
```

EXERCISE 2B- ANOVA AND SAMPLE SIZE

Calculating N with pwr package
(For effect size, we need grand average)

To do in class …

Recalculate exercise 2a fixing this sample size

POWER AND SAMPLE SIZE

Exercise 3:

- Consider the following clinical trial:

The NEW ENGLAND JOURNAL of MEDICINE

CURRENT ISSUE SPECIALTIES TOPICS

ORIGINAL ARTICLE

A Randomized Trial of Intensive versus Standard Blood-Pressure Control

This article has been corrected. VIEW THE CORRECTION

Author: The SPRINT Research Group* Author Info & Affiliations

Published November 26, 2015 | N Engl J Med 2015;373:2103-2116 | DOI: 10.1056/NEJMoa1511939
VOL. 373 NO. 22 | Copyright © 2015

Abstract

BACKGROUND

The most appropriate targets for systolic blood pressure to reduce cardiovascular morbidity and mortality among persons without diabetes remain uncertain.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4156910/pdf/nihms593625.pdf

https://www.nejm.org/doi/full/10.1056/NEJMoa1511939

POWER AND SAMPLE SIZE

Exercise 3:

SPRINT STUDY

A randomized trial of intensive versus standard blood-pressure control.

Multicenter, Randomized, Placebo-Controlled Trial

Rationale : In patients WITHOUT a history of diabetes or stroke, does intensive blood pressure control improve outcomes?

Is intensive BP control (SBP <120mmHg) better than standard control (SBP 135 to 139mmHg) ?

n=9,361

- > 1 CVD risk factor
- SBP ranging from 130-180mmHg
- Age > 50 years

INTENSIVE SBP <120mmHg

5.2% Primary Outcome*

STANDARD SBP 135-139mmHg

6.8% Primary Outcome*

Primary Outcome : First acute coronary syndrome, stroke, heart failure or CV death*
Hazard Ratio of Intensive vs Standard, 0.75, 95% CI 0.64-0.89 ; p <0.001

3.3% All-cause mortality

4.5% All-cause mortality

All-cause mortality
Hazard Ratio of Intensive vs Standard, 0.73, 95% CI 0.60-0.90 ; p = 0.003

CLINICAL PEARLS

Intensive blood pressure control improves cardio-vascular outcomes in patients at a high risk for CVD. There was a modest risk of serious adverse events

ACC/AHA REVISED DEFINITIONS (2017)

- Normal Blood Pressure, <120/80mmHg
- Elevated Blood Pressure, 120-129/<80mmHg
- HTN stage I, 130-139/80-89mmHg
- HTN stage II, >140/90

BLOOD PRESSURE GOAL IN DIABETES

130/80

ENDOCONSULT

N Engl J Med 2015; 373:2103-2116

POWER AND SAMPLE SIZE

Exercise 3:

- Consider the following clinical trial:

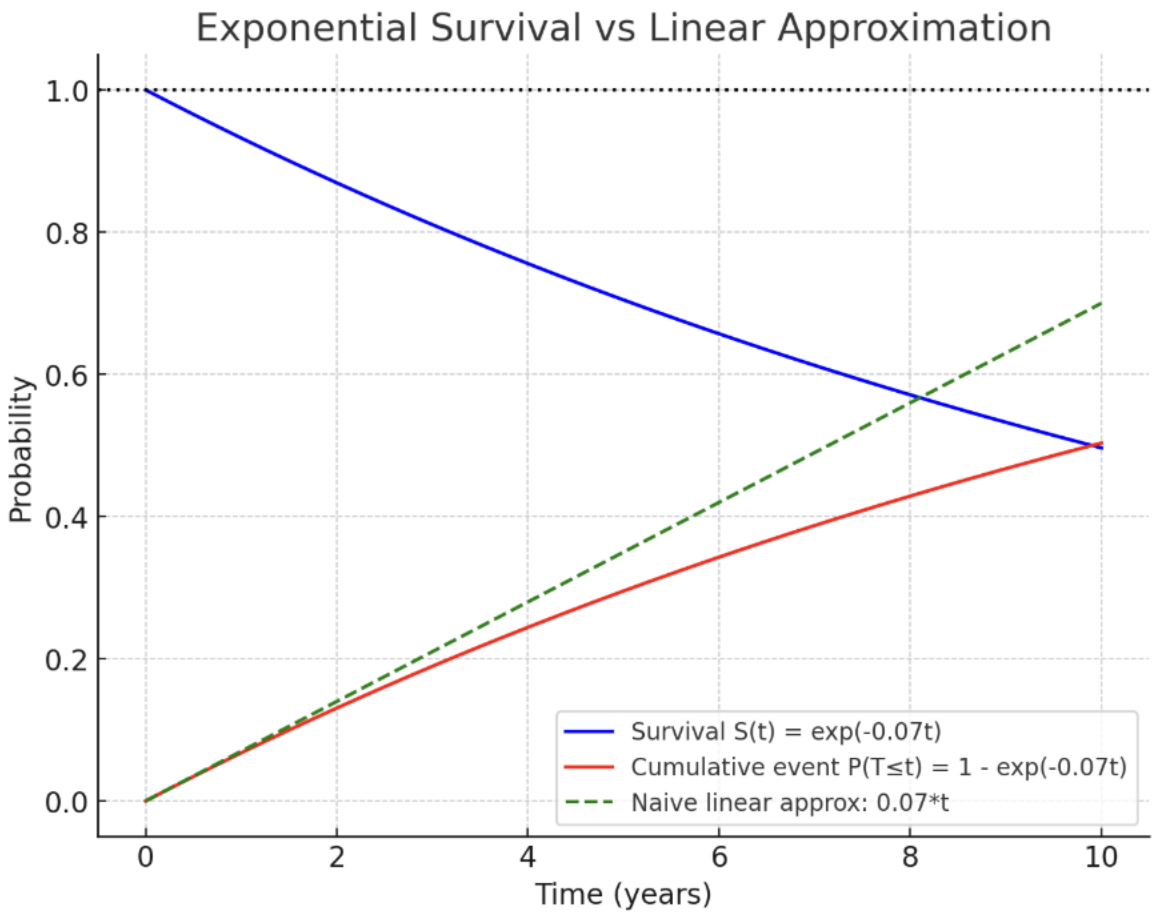
  We assumed a 2-year recruitment period, a maximum follow-up of 6 years. **With the targeted 9250 participants, we calculated 88.7% power to detect a 20% effect of the composite outcome, assuming a 7%/yr event rate in the Standard Group.**

  Are they correct? How the solution changes if they only eventually recruited 2960 participants? Assume in both cases 50% of cases for each group.

POWER AND SAMPLE SIZE

Exercise 3:

- Consider the following clinical trial:



POWER AND SAMPLE SIZE

Balancing sample size, effect size and power is critical to good study design. We begin by setting the values of type I error (α) and power ($1 - \beta$) to be statistically adequate: traditionally 0.05 and 0.80, respectively. We then determine n on the basis of the smallest effect we wish to measure. If the required sample size is too large, we may need to reassess our objectives or more tightly control the experimental conditions to reduce the variance. Use the interactive graphs in **Supplementary Table 1** to explore power calculations.

Krzywinski and Altman, 2013, Nature Methods

POWER AND SAMPLE SIZE

Some Real Examples:

- Depending on the case we may use different approaches (proportion tests, survival analysis, ROC curves, etc)

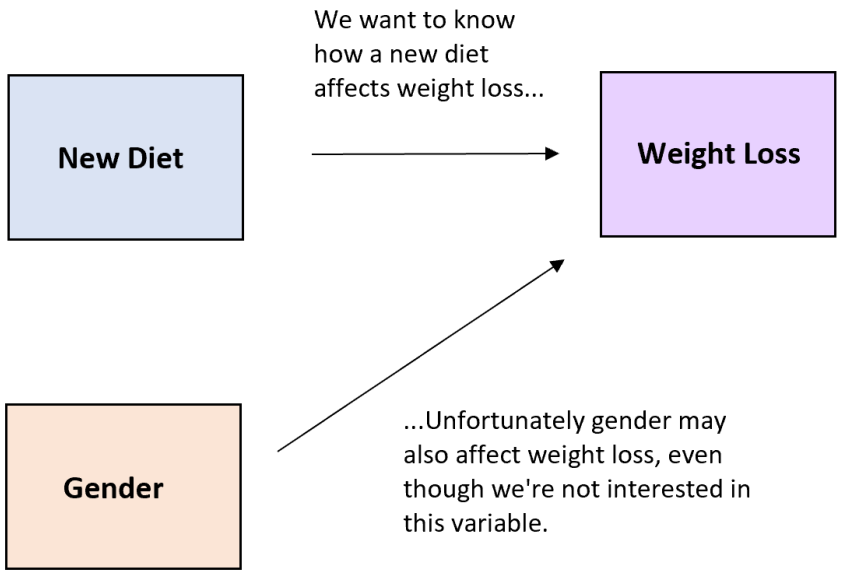
  - *https://bmcwomenshealth.biomedcentral.com/articles/10.1186/1472-6874-13-3*
  - https://www.sciencedirect.com/science/article/abs/pii/S0959652615012366?via%3Dihub
  - https://www.sciencedirect.com/science/article/abs/pii/S0924013606008272?via%3Dihub
  - *https://www.ncbi.nlm.nih.gov/pubmed/21450436*
  - *http://www.thelancet.com/journals/lancet/article/PIIS0140-6736(17)32346-2/fulltext*
  - ….

SOURCES OF VARIATION

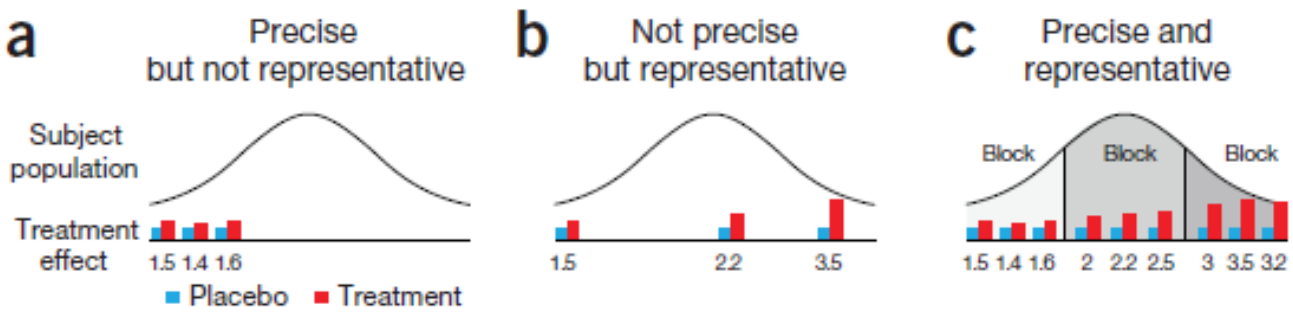
- Variation limits our ability to detect effects by reducing the power of tests
- Technical variability (measurement devices)
- **Complexity**. Different factors that affects the response variable (subgroups) when trying to generalize the results to the population of interest.

SOURCES OF VARIATION

- Gender effect over diet / opposite effect. Real examples!



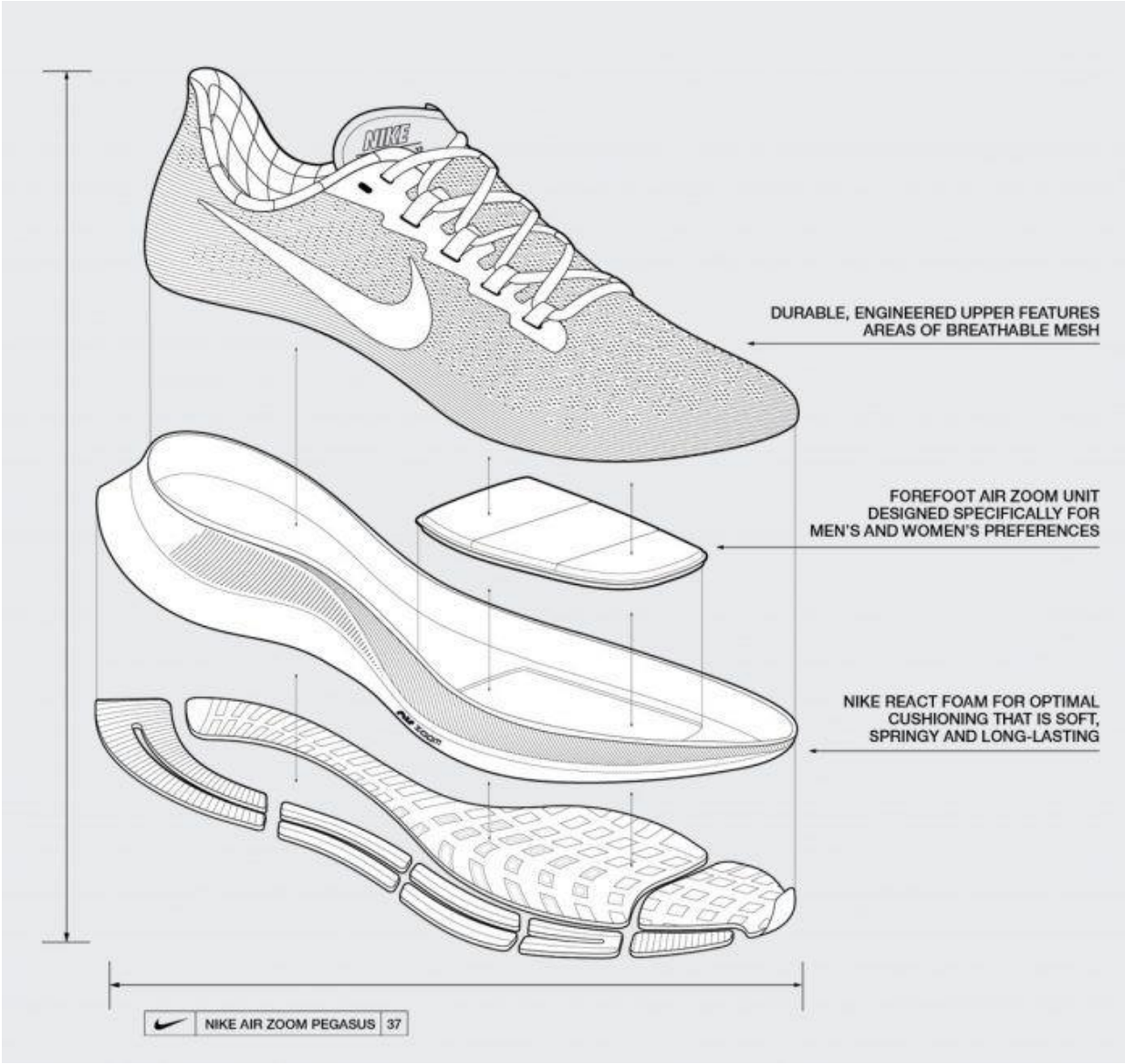
SOURCES OF VARIATION



Variability is inevitable in experiments owing to both biological and technical effects. Whereas technical variability should be tightly controlled to enhance the internal validity of the results, some types of biological variability need to be maintained to allow generalization of the results to the population of interest. Experimental control, randomization, blocking and replication are the tools that allow replicable and meaningful results to be obtained in the face of variability.

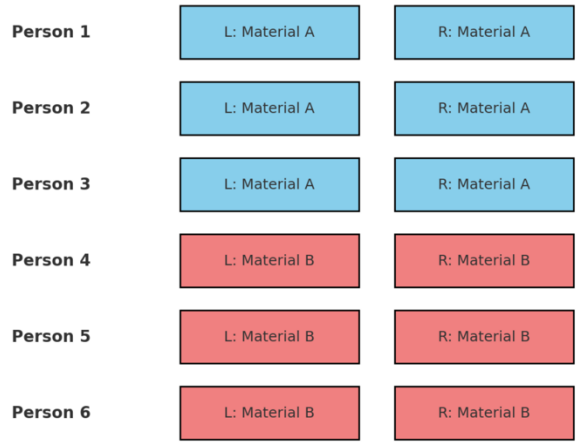
Altman and Krzywinski , 2015, Nature Methods

BEYOND THE LIMITS


NIKE
DURABLE, ENGINEERED UPPER FEATURES
AREAS OF BREATHABLE MESH
FOREFOOT AIR ZOOM UNIT
DESIGNED SPECIFICALLY FOR
MEN'S AND WOMEN'S PREFERENCES
NIKE REACT FOAM FOR OPTIMAL
CUSHIONING THAT IS SOFT,
SPRINGY AND LONG-LASTING
NIKE AIR ZOOM PEGASUS 37

SOURCES OF VARIATION

- Variation limits our ability to detect effects by reducing the power of tests
- Technical variability.
- **Complexity**. Different factors that affects the response variable when trying to generalize the results to the population of interest.
- **Example:** comparing shoe wear with different materials (A,B). Can you think in a good experimental design? Assume we have six individuals.

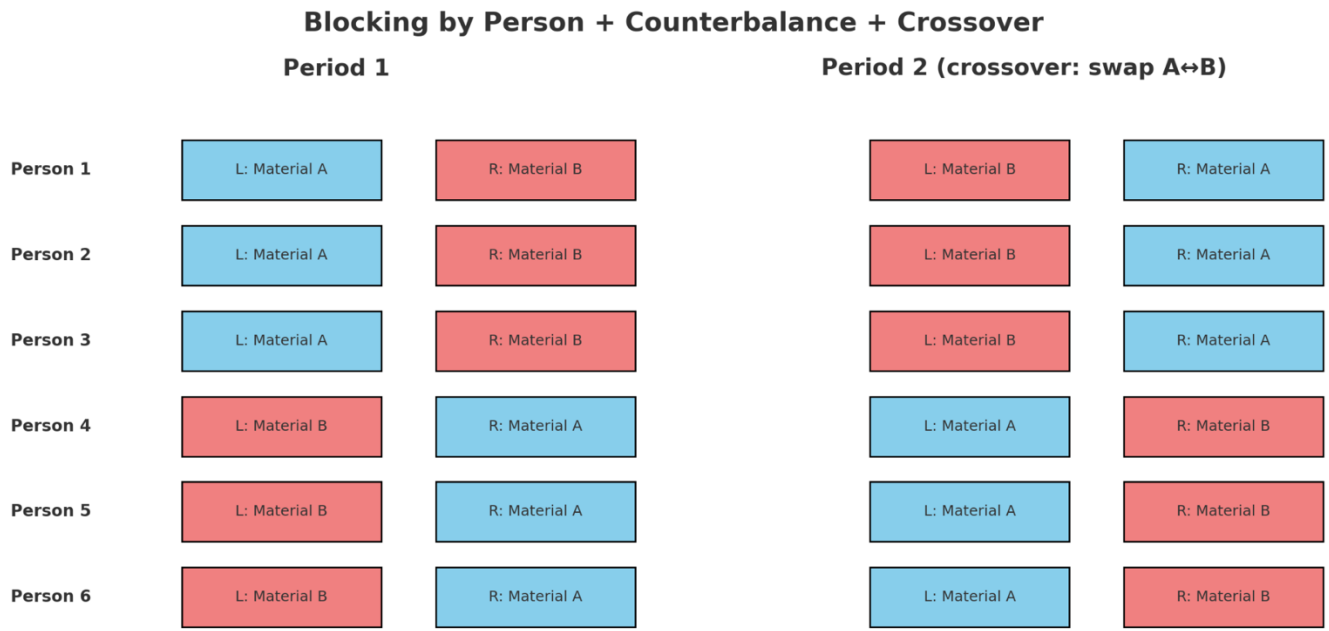


3 replicates for each individual

Is this a good design? Analyze and propose improvements

SOURCES OF VARIATION

A much better design:



- We **block** the variability across individuals
- We cannot block the effect of right and left shoe. We do **random** assignment.
- 3 **replicates** we increase precision.
- Crossover

SOURCES OF VARIATION

- **Blocking**: portion of the experimental material that is expected to be more homogenous than the aggregate
  - Example: **Paired samples**
  - Runs made close together in time or space are likely to be more similar than runs made further apart.
- **Randomization**: uncontrolled noise is homogenized between different groups
- **Replicates**. In our example, should we collect data from more individuals or technical replicates?

SOURCE OF VARIATION

Simple Block Design:

- Factor A, called treatment, of interest
- Factor B, called blocks, known source of variability that we need to control
- Our interest is comparing the level of treatments
- 2-way ANOVA

Exercise 4:
Generate **N=50** random values for 3 different groups, which are normally distributed as follows: **N(μ_1,1), N(μ_2,1)** and **N(μ_3,1)** , μ_1=1; μ_2=2; μ_3=3. Data is generated into two different blocks. One of them strongly adds a bias into the measure, modelled here as normal distribution **N(μ_B,1), μ_B**=10**.**
Generate the dataset and conduct ANOVA taking into account the blocks.

Block – gender
Diets – weight gain after diet

EXERCISE 4

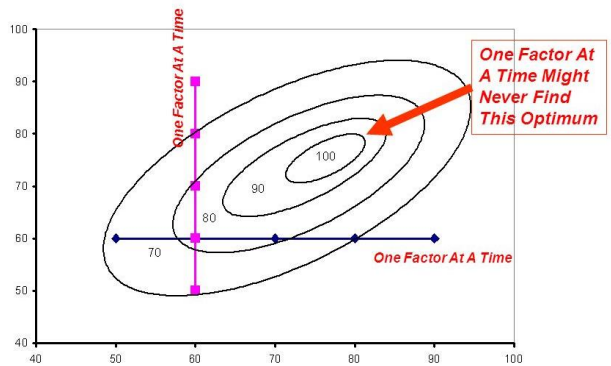
```
# Clean the workspace
rm(list = ls()); graphics.off(); set.seed(500)

# Input Data
n <- 50; u1 <-1; u2 <-2; u3 <-3; sigma<-1; k <- 3; ub <- 10

# Generate factors
f <- c("T1", "T2", "T3")
tm <- gl(k,1, n*k, factor (f))
blocks <- gl(2, 3*n/2, labels=c("Block1", "Block2"))

# Generate output values
b <- c(rep(0,n/2),rep(1,n/2))
t1 <- rnorm(n, u1,sigma) + b*rnorm(n, ub, sigma)
t2 <- rnorm(n, u2,sigma) + b*rnorm(n, ub, sigma)
t3 <- rnorm(n, u3,sigma) + b*rnorm(n, ub, sigma)
y <- matrix (c(t1,t2, t3), nrow=n, ncol=k)
y <- c(t(y))
```

EXERCISE 4

```
# Integrate data into data frame

data1 <- data.frame(tm, blocks,y)

# Anova with blocks
av <-  aov(y ~ tm + blocks, projections=TRUE, data=data1)
summary(av)

# Anova without blocks
av <-  aov(y ~ tm, projections=TRUE, data=data1)
summary(av)
```

DESIGN OF EXPERIMENTS (DOE)

- **Statistical techniques** to define a **rational experimental plan** in order to identify and quantify the **causes of an effect** and its more **relevant variables**.
- **Basic strategies**:
  - One-factor-at-a-time (OFAT) method: fix all values minus one



DESIGN OF EXPERIMENTS (DOE)

DESIGN OF EXPERIMENTS (DOE)

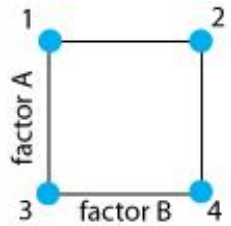
1. Raw Materials & Procurement (iron, titanium)
2. Casting & Forging
3. Machining of Components
4. Heat Treatment & Surface Finishing (Temperature)
5. Sub-Assembly Production
6. Final Assembly (machine)
7. Electronics & Control Systems Integration (one software or other)

DESIGN OF EXPERIMENTS (DOE)

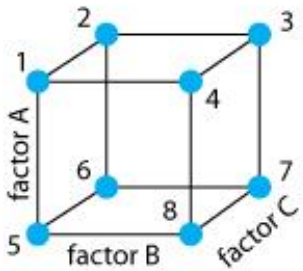
- **Design of Experiments (DoE):** mathematical framework for optimal design of experiments.
- Identification of the **interactions among different experimental factors** in order to obtain optimal combinations.
- Objectives:
  - Identify the most relevant factors: **Complete and Fractional Factorial designs (high and low)**
  - Identify levels of factors that optimize the response: **Response Surface Methods (Central Composite Design) fine tunning**
  - Identify levels of factors that minimize variability due to experimental noise (**robustness**)

FACTORIAL DESIGNS

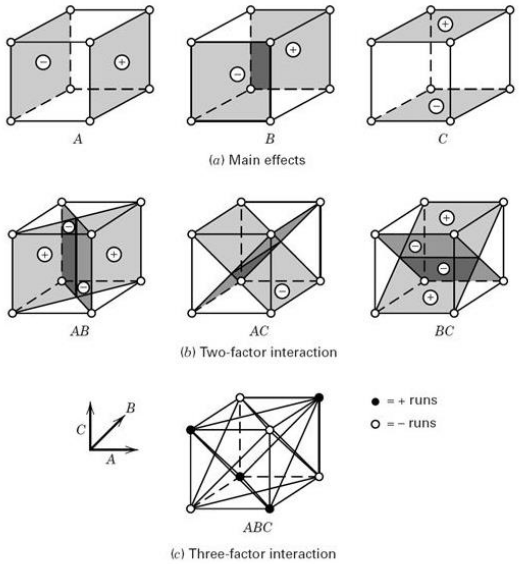
- Selection of a fixed number of levels for each factor (k in total) considered and run experiments in all possible combinations.
- Two levels for each factor: $\mathbf{2^k}$
- **Main effects** and **interactions**.
- Promising direction for further experimentation



	factor levels	
trial	A	B
1	+	−
2	+	+
3	−	−
4	−	+



	factor levels		
trial	A	B	C
1	+	−	−
2	+	−	+
3	+	+	+
4	+	+	−
5	−	−	−
6	−	−	+
7	−	+	+
8	−	+	−



FACTORIAL DESIGNS (INTERACTIONS)

2 Factor Factorial Design

Trial	A	B	AB
1	+	-	-
2	+	+	+
3	-	-	+
4	-	+	-

FACTORIAL DESIGNS

3 Factor Factorial Design

Trial	A	B	C	AB	AC	BC	ABC
1	-	-	-	+	+	+	+
2	-	-	+	+	-	-	-
3	-	+	-	-	+	-	-
4	-	+	+	-	-	+	-
5	+	-	-	-	-	+	-
6	+	-	+	-	+	-	-
7	+	+	-	+	-	-	-
8	+	+	+	+	+	+	+

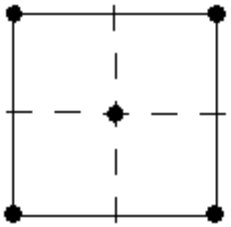
FACTORIAL DESIGNS

- **Resolution**.
  - **Resolution III**: main effects confounded with doble interactions.
  - **Resolution IV**: doble interactions confounded
  - **Resolution V**: main effects and doble interactions not confounded
  - **Complete resolution**: no effects and interactions confounded.

	Factors						
Run	A	B	C	D	E	F	G
1	−	−	+	−	+	+	−
2	−	−	+	+	−	−	+
3	−	+	−	−	+	−	+
4	−	+	−	+	−	+	−
5	+	−	−	−	−	+	+
6	+	−	−	+	+	−	−
7	+	+	+	−	−	−	−
8	+	+	+	+	+	+	+

Table 2. An eight run two-level fractional factorial experiment for varying seven factors. One line in the table corresponds to a single run. A minus sign in a column means that the factor listed above the column should be at its low level. A plus sign indicates that the factor should be at its high level in that particular run.

- **Center points**.
  - Sequencial strategy: non-linear effects
  - Variance prediction in the case replicates are not possible.



FACTORIAL DESIGNS

Shoe-wear idea, for the material generation.

A: Cure temperature (Low / High)
B: UV dose (Low / High)
C: Line speed (Slow / Fast)
D: Additive package (Old / New)

Resolution IV

Run	A	B	C	D (=A·B·C)
1	−1	−1	−1	−1
2	+1	−1	−1	+1
3	−1	+1	−1	+1
4	+1	+1	−1	−1
5	−1	−1	+1	+1
6	+1	−1	+1	−1
7	−1	+1	+1	−1
8	+1	+1	+1	+1

FRACTIONAL FACTORIAL DESIGNS

- Simplest case, $\mathbf{2_x^{k-p}}$: k factors of 2 levels using $\mathbf{2^{k-p}}$ runs of resolution x
- **Screening purposes.** Main effects only.
- More than double interactions are typically neglected.
- Consider the $\mathbf{2_{III}^{7-4}}$ design.
- We can learn from these designs the **following steps for experimentation**: factors to be deleted, new levels, etc.
- **Plackett and Burman (PB) Designs (Resolution III)**:
  - Multiple of 4: n=12,20,24…
  - Projective properties
  - *k*+1 runs to determine *k* factors

*8 factors – Full resolution (**256**) or PB III (**9**)*
Cheaper design!

	Factors						
Run	A	B	C	D	E	F	G
1	−	−	+	−	+	+	−
2	−	−	+	+	−	−	+
3	−	+	−	−	+	−	+
4	−	+	−	+	−	+	−
5	+	−	−	−	−	+	+
6	+	−	−	+	+	−	−
7	+	+	+	−	−	−	−
8	+	+	+	+	+	+	+

Table 2. An eight run two-level fractional factorial experiment for varying seven factors. One line in the table corresponds to a single run. A minus sign in a column means that the factor listed above the column should be at its low level. A plus sign indicates that the factor should be at its high level in that particular run.

Plackett - Burman Design

Factors:	16	Replicates:	1
Base runs:	20	Total runs:	20
Base blocks:	1	Total blocks:	1

Design Table

Run	Blk	A	B	C	D	E	F	G	H	J	K	L	M	N	O	P	Q
1	1	+	-	+	+	-	-	-	-	+	-	+	-	+	+	+	+
2	1	+	+	-	+	+	-	-	-	-	+	-	+	-	+	+	+
3	1	-	+	+	-	+	+	-	-	-	-	+	-	+	-	+	+
4	1	-	-	+	+	-	+	+	-	-	-	-	+	-	+	-	+
5	1	+	-	-	+	+	-	+	+	-	-	-	-	+	-	+	-
6	1	+	+	-	-	+	+	-	+	+	-	-	-	-	+	-	+
7	1	+	+	+	-	-	+	+	-	+	+	-	-	-	-	+	-
8	1	+	+	+	+	-	-	+	+	-	+	+	-	-	-	-	+
9	1	-	+	+	+	+	-	-	+	+	-	+	+	-	-	-	-
10	1	+	-	+	+	+	+	-	-	+	+	-	+	+	-	-	-
11	1	-	+	-	+	+	+	+	-	-	+	+	-	+	+	-	-
12	1	+	-	+	-	+	+	+	+	-	-	+	+	-	+	+	-
13	1	-	+	-	+	-	+	+	+	+	-	-	+	+	-	+	+
14	1	-	-	+	-	+	-	+	+	+	+	-	-	+	+	-	+
15	1	-	-	-	+	-	+	-	+	+	+	+	-	-	+	+	-
16	1	-	-	-	-	+	-	+	-	+	+	+	+	-	-	+	+
17	1	+	-	-	-	-	+	-	+	-	+	+	+	+	-	-	+
18	1	+	+	-	-	-	-	+	-	+	-	+	+	+	+	-	-
19	1	-	+	+	-	-	-	-	+	-	+	-	+	+	+	+	-
20	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

EXERCISE 5

a) Assume that you have 2^3 factorial design with two different replicates (16 runs), being: y = [59,74,50,69,50,81,46,79, 61,70,58,67,54,85,44,81]. Calculate the design matrix, main effects and interactions using the package **FrF2** and function ***lm***.

b) Construct a 2^{7-4} factorial fractional design using package ***FrF2*** in R. Analyze the main effects of this design for the following response [69, 52, 60, 83, 71, 50, 59, 88].

c) A research group wants to test whether the presence/absence of 5 nutrients affects the levels of glucose in blood. At this stage of the study, only main effects and double interactions want to be assessed. Establish an appropriate experimental design to answer this question. Due to the technical variability, three replicates per experiment will be required. Consider using the minimum number of experiments. (<u>Resolution V</u>).

EXERCISE 5A

```
# Clean the workspace
rm(list = ls()); graphics.off(); set.seed(500)

# Load libray FrF2
library(FrF2)

# Generate response and design matrix
y <- c(59,74,50,69,50,81,46,79,
      61,70,58,67,54,85,44,81)
design1 <- FrF2(nruns=8, nfactors=3,  replications=2,
randomize=FALSE)
design1 <- add.response(design1,y)

# Conduct linear model
M1 <- lm(design1, degree=3)
summary(M1)
```

EXERCISE 5B

```
# Clean the workspace
rm(list = ls()); graphics.off(); set.seed(500)

# Load libray FrF2
library(FrF2)

# Generate response and design matrix

y <- c(69, 52, 60, 83, 71, 50, 59, 88)
design2 <- FrF2(nruns= 8,nfactor=7, randomize=FALSE)
design2 <- add.response(design2,y)

# Conduct linear model
M2 <- lm(design2, degree=1)
summary(M2)

## saturated, but estimation…
```

- $\mathbf{2}_{x}^{k-p}$: k factors of 2 levels using $\mathbf{2}^{k-p}$ runs of resolution x

FACTORIAL DESIGNS

- **Resolution**.
  - **Resolution III**: main effects confounded with doble interactions.
  - **Resolution IV**: doble interactions confounded
  - **Resolution V**: main effects and doble interactions not confounded
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	Factors						
Run	A	B	C	D	E	F	G
1	−	−	+	−	+	+	−
2	−	−	+	+	−	−	+
3	−	+	−	−	+	−	+
4	−	+	−	+	−	+	−
5	+	−	−	−	−	+	+
6	+	−	−	+	+	−	−
7	+	+	+	−	−	−	−
8	+	+	+	+	+	+	+

Table 2. An eight run two-level fractional factorial experiment for varying seven factors. One line in the table corresponds to a single run. A minus sign in a column means that the factor listed above the column should be at its low level. A plus sign indicates that the factor should be at its high level in that particular run.

- **Center points**.
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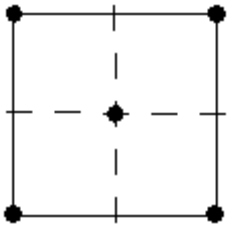

EXERCISE 5C

```
# Clean the workspace
rm(list = ls()); graphics.off(); set.seed(500)
```

To do in class …

REFERENCES

- *Introduction to sample size determination and power analysis for clinical trials*, John M. Lachin, Controlled Clinical Trial, 1981
- *Power and sample size*, Martin Krzywinski & Naomi Altman, 2013, Nature Methods
- *Experimental Designs*, William Gemmell Cochran and Gertrude M. Cox
- *Statistics for experimenters*, George E.P. Box, J. Stuart Hunter and William G. Hunter.
- *Power failure: why small sample size undermines the reliability of neuroscience*. Button, K. S. et al (2013). Nature Reviews Neuroscience

ESTADÍSTICA Y DISEÑO EXPERIMENTAL

EXPERMIENTAL DESIGN

Rafael Valdés Mas
rvaldesmas@unav.es